# Binomial Theorem: Questions and Full Solutions

## Question 01

**English**

Let $a$ be a real number. In the expansion of $\left(a^3x^3 - 3a^2x + \frac{3a}{x} - \frac{1}{x^3}\right)^4$, $x \neq 0$, in powers of $x$, the coefficient of $x^{-10}$ is $-6$. Then the term independent of $x$ in this expansion is equal to

(A) $\frac{231}{8}$

(B) $-\frac{231}{16}$

(C) $\frac{231}{16}$

(D) $\frac{495}{16}$

Only one option is correct.

**हिन्दी**

माना कि $a$ एक वास्तविक संख्या है। $x$ की घातों में $\left(a^3x^3 - 3a^2x + \frac{3a}{x} - \frac{1}{x^3}\right)^4$, $x \neq 0$, के प्रसार में $x^{-10}$ का गुणांक $-6$ है। तब इस प्रसार में $x$ से स्वतंत्र पद बराबर है

(A) $\frac{231}{8}$

(B) $-\frac{231}{16}$

(C) $\frac{231}{16}$

(D) $\frac{495}{16}$

केवल एक विकल्प सही है।

**Solution**

Expand the cube $\left(ax - \frac{1}{x}\right)^3 = a^3x^3 - 3(ax)^2 \cdot \frac{1}{x} + 3(ax) \cdot \frac{1}{x^2} - \frac{1}{x^3} = a^3x^3 - 3a^2x + \frac{3a}{x} - \frac{1}{x^3}$.

So the given four-term base is exactly $\left(ax - \frac{1}{x}\right)^3$, and its fourth power is $\left(ax - \frac{1}{x}\right)^{12}$. This turns a multinomial expansion into a binomial one.

General term: $T_{r+1} = \binom{12}{r}(ax)^{12-r}\left(-\frac{1}{x}\right)^r = \binom{12}{r}a^{12-r}(-1)^r x^{12-2r}$, $r = 0, 1, \ldots, 12$.

Coefficient of $x^{-10}$: $12 - 2r = -10$ gives $r = 11$, and the coefficient is $\binom{12}{11}a^1(-1)^{11} = -12a$.

Given $-12a = -6$, so $a = \frac{1}{2}$ (a linear equation, so $a$ is unique).

Term independent of $x$: $12 - 2r = 0$ gives $r = 6$, and the term is $\binom{12}{6}a^6(-1)^6 = 924 \cdot \frac{1}{64} = \frac{231}{16}$.

(Option (D) is $\binom{12}{4}a^4$, the result of using $r = 4$; option (B) has the wrong sign.)

**Correct Option: (C)**

## Question 02

**English**

The expression $\left(\sqrt{3}+\sqrt[3]{5}\right)^{50}+\left(\sqrt{3}-\sqrt[3]{5}\right)^{50}$ is expanded by the binomial theorem and the like terms are combined. Then the number of irrational terms in the resulting expression is

(A) 17
(B) 42
(C) 18
(D) 26

Only one option is correct.

**हिन्दी**

व्यंजक $\left(\sqrt{3}+\sqrt[3]{5}\right)^{50}+\left(\sqrt{3}-\sqrt[3]{5}\right)^{50}$ का द्विपद प्रमेय द्वारा प्रसार करके समान पदों को मिलाया जाता है। तब प्राप्त व्यंजक में अपरिमेय पदों की संख्या है

(A) 17
(B) 42
(C) 18
(D) 26

केवल एक विकल्प सही है।

**Solution**

Expansion: $\left(\sqrt{3}\pm\sqrt[3]{5}\right)^{50}=\sum_{r=0}^{50}\binom{50}{r}(\pm1)^r 3^{\frac{50-r}{2}}5^{\frac{r}{3}}$.

Adding the two expansions, the terms with odd $r$ cancel and those with even $r$ appear with coefficient $2\binom{50}{r}$. So after combining, the expression has one term for each even $r$ in $0,2,\ldots,50$: 26 terms.

A surviving term $2\binom{50}{r}3^{\frac{50-r}{2}}5^{\frac{r}{3}}$ is rational exactly when both exponents are integers. Since $r$ is even, $\frac{50-r}{2}$ is an integer; $\frac{r}{3}$ is an integer iff $3 \mid r$.

So rational terms correspond to even multiples of 3, i.e. multiples of 6: $r=0,6,12,\ldots,48$, which is 9 values.

The remaining $26-9=17$ terms contain $5^{1/3}$ or $5^{2/3}$ and are irrational (distinct terms have different powers, so they do not merge into rational terms).

(Counting in a single expansion would give $51-9=42$, option (B); the cancellation of odd $r$ is the point of the question.)

**Correct Option: (A)**

## Question 03

**English**

Let $m$ and $n$ be non-negative integers such that, in the expansion of $(1+x)^m(1-x)^n$, the coefficient of $x$ is 4 and the coefficient of $x^2$ is 2. Then the coefficient of $x^3$ in this expansion is

(A) 12
(B) $-12$
(C) $-8$
(D) $-16$

Only one option is correct.

**हिन्दी**

माना कि $m$ तथा $n$ ऐसे अऋणात्मक पूर्णांक हैं कि $(1+x)^m(1-x)^n$ के प्रसार में $x$ का गुणांक 4 तथा $x^2$ का गुणांक 2 है। तब इस प्रसार में $x^3$ का गुणांक है

(A) 12
(B) $-12$
(C) $-8$
(D) $-16$

केवल एक विकल्प सही है।

**Solution**

$(1+x)^m = 1 + mx + \binom{m}{2}x^2 + \ldots$ and $(1-x)^n = 1 - nx + \binom{n}{2}x^2 - \ldots$

Coefficient of $x$ in the product: $m - n$. Given $m - n = 4$.

Coefficient of $x^2$: $\binom{m}{2} - mn + \binom{n}{2} = \dfrac{m^2 - m - 2mn + n^2 - n}{2} = \dfrac{(m-n)^2 - (m+n)}{2}$.

Given it equals 2: $16 - (m+n) = 4 \Rightarrow m + n = 12$. With $m - n = 4$: $m = 8$, $n = 4$ (unique).

Now $(1+x)^8(1-x)^4 = [(1+x)(1-x)]^4\,(1+x)^4 = (1-x^2)^4(1+x)^4$.

$(1-x^2)^4 = 1 - 4x^2 + \ldots$ and $(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$.

$x^3$ arises as $1 \cdot 4x^3$ and $(-4x^2)(4x)$: coefficient $4 - 16 = -12$.

Check by the direct formula: $\binom{8}{3} - \binom{8}{2} \cdot 4 + 8\binom{4}{2} - \binom{4}{3} = 56 - 112 + 48 - 4 = -12$.

**Correct Option: (B)**

## Question 04

**English**

Let $n$ be a positive integer such that $\sum_{r=0}^{n} \frac{\binom{n}{r}}{r+1} = \frac{1023}{10}$. Then $\sum_{r=1}^{n} r^2 \binom{n}{r}$ is equal to

(A) 9216
(B) 2304
(C) 12800
(D) 11520

Only one option is correct.

**हिन्दी**

माना कि $n$ एक ऐसा धनात्मक पूर्णांक है कि $\sum_{r=0}^{n} \frac{\binom{n}{r}}{r+1} = \frac{1023}{10}$ है। तब $\sum_{r=1}^{n} r^2 \binom{n}{r}$ बराबर है

(A) 9216
(B) 2304
(C) 12800
(D) 11520

केवल एक विकल्प सही है।

**Solution**

$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r$. Integrating from 0 to 1: $\frac{2^{n+1}-1}{n+1} = \sum_{r=0}^{n} \frac{\binom{n}{r}}{r+1}$.

So $\frac{2^{n+1}-1}{n+1} = \frac{1023}{10}$. For $n = 9$: $\frac{1023}{10}$, which matches.

Uniqueness: $g(n) = \frac{2^{n+1}-1}{n+1}$ satisfies $g(n+1) > g(n)$ for $n \geq 1$ (the numerator roughly doubles while the denominator grows by 1), so $n = 9$ is the only solution.

For $\sum r^2 \binom{n}{r}$: differentiate $(1+x)^n$ to get $n(1+x)^{n-1} = \sum r \binom{n}{r} x^{r-1}$; multiply by $x$: $nx(1+x)^{n-1} = \sum r \binom{n}{r} x^r$.

Differentiate again: $n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2} = \sum r^2 \binom{n}{r} x^{r-1}$.

Put $x = 1$: $\sum r^2 \binom{n}{r} = n2^{n-1} + n(n-1)2^{n-2} = n(n+1)2^{n-2}$.

With $n = 9$: $9 \cdot 10 \cdot 128 = 11520$.

(Option (A) is $\sum r(r-1)\binom{9}{r} = 9 \cdot 8 \cdot 2^7$ and option (B) is $\sum r \binom{9}{r} = 9 \cdot 2^8$.)

**Correct Option: (D)**

## Question 05

**English**

For each integer $k$ with $0 \le k \le 10$, let $T_k = \sum_{r=0}^{10-k} (-1)^r \binom{10}{r} \binom{10}{r+k}$. Then $\sum_{k=0}^{10} T_k$ is equal to

(A) $-126$
(B) 126
(C) 0
(D) $-252$

Only one option is correct.

**हिन्दी**

प्रत्येक पूर्णांक $k$, $0 \le k \le 10$, के लिए माना $T_k = \sum_{r=0}^{10-k} (-1)^r \binom{10}{r} \binom{10}{r+k}$ है। तब $\sum_{k=0}^{10} T_k$ बराबर है

(A) $-126$
(B) 126
(C) 0
(D) $-252$

केवल एक विकल्प सही है।

**Solution**

Write $\binom{10}{r+k} = \binom{10}{10-k-r}$. Then $T_k = \sum_r (-1)^r \binom{10}{r} \binom{10}{(10-k)-r}$, which is the coefficient of $x^{10-k}$ in the product of $(1-x)^{10} = \sum (-1)^r \binom{10}{r} x^r$ and $(1+x)^{10} = \sum \binom{10}{s} x^s$.

That product is $(1-x^2)^{10} = \sum_{j=0}^{10} (-1)^j \binom{10}{j} x^{2j}$.

Hence $T_k = 0$ when $10-k$ is odd, and $T_k = (-1)^j \binom{10}{j}$ when $10-k = 2j$. As $k$ runs over $0, \ldots, 10$, $10-k$ runs over $0, \ldots, 10$, so the even values give $j = 0, 1, \ldots, 5$.

So $\sum_{k=0}^{10} T_k = \sum_{j=0}^{5} (-1)^j \binom{10}{j} = 1 - 10 + 45 - 120 + 210 - 252$.

Using $\binom{n}{j} = \binom{n-1}{j} + \binom{n-1}{j-1}$ the partial alternating sum telescopes: $\sum_{j=0}^{p} (-1)^j \binom{n}{j} = (-1)^p \binom{n-1}{p}$.

With $n = 10$, $p = 5$: $-\binom{9}{5} = -126$ (direct check: $1 - 10 + 45 - 120 + 210 - 252 = -126$).

(Summing over all coefficients of $(1-x^2)^{10}$ instead, i.e. putting $x = 1$, would give 0, option (C).)

**Correct Option: (A)**

## Question 06

**English**

Here $\log_e$ denotes the natural logarithm. The value of $\lim\limits_{x\to 0}\dfrac{(e^x-1)\log_e(1+x)-x^2}{x^4}$ is

(A) $\dfrac{1}{12}$
(B) $\dfrac{1}{4}$
(C) $0$
(D) $\dfrac{1}{2}$

Only one option is correct.

**हिन्दी**

यहाँ $\log_e$ प्राकृतिक लघुगणक को दर्शाता है। $\lim\limits_{x\to 0}\dfrac{(e^x-1)\log_e(1+x)-x^2}{x^4}$ का मान है

(A) $\dfrac{1}{12}$
(B) $\dfrac{1}{4}$
(C) $0$
(D) $\dfrac{1}{2}$

केवल एक विकल्प सही है।

**Solution**

Use the series $e^x-1=x+\dfrac{x^2}{2}+\dfrac{x^3}{6}+O(x^4)$ and $\log_e(1+x)=x-\dfrac{x^2}{2}+\dfrac{x^3}{3}+O(x^4)$.

Multiply. $x^2$: $x\cdot x=x^2$.

$x^3$: $x\cdot\left(-\dfrac{x^2}{2}\right)+\dfrac{x^2}{2}\cdot x=0$.

$x^4$: $x\cdot\dfrac{x^3}{3}+\dfrac{x^2}{2}\cdot\left(-\dfrac{x^2}{2}\right)+\dfrac{x^3}{6}\cdot x=\left(\dfrac{1}{3}-\dfrac{1}{4}+\dfrac{1}{6}\right)x^4=\dfrac{x^4}{4}$.

So the numerator is $\dfrac{x^4}{4}+O(x^5)$, and the limit is $\dfrac{1}{4}$.

(Stopping at the $x^3$ term would wrongly suggest the limit 0, option (C); the series must be carried to the fourth order.)

**Correct Option: (B)**

## Question 07

**English**

The remainder when $3^{5^{100}}$ is divided by 77 is

(A) 3
(B) 1
(C) 34
(D) 45

Only one option is correct.

**हिन्दी**

$3^{5^{100}}$ को 77 से भाग देने पर शेषफल है

(A) 3
(B) 1
(C) 34
(D) 45

केवल एक विकल्प सही है।

**Solution**

Since $\gcd(3, 77) = 1$ and $\phi(77) = \phi(7)\phi(11) = 6 \cdot 10 = 60$, Euler's theorem gives $3^{60} \equiv 1 \pmod{77}$. So only $5^{100} \bmod 60$ matters.

$5^2 = 25$ and $5^3 = 125 = 2 \cdot 60 + 5$, so $5^3 \equiv 5$, and then $5^k \equiv 5$ for odd $k$, $5^k \equiv 25$ for even $k \geq 2$ (modulo 60). Hence $5^{100} \equiv 25$, and $3^{5^{100}} \equiv 3^{25} \pmod{77}$.

Modulo 7 (Fermat, 7 prime): $3^6 \equiv 1$, $25 = 4 \cdot 6 + 1$, so $3^{25} \equiv 3$.

Modulo 11 (Fermat, 11 prime): $3^{10} \equiv 1$; in fact $3^5 = 243 = 22 \cdot 11 + 1 \equiv 1$, so $3^{25} = (3^5)^5 \equiv 1$.

Find $N \in \{0, \ldots, 76\}$ with $N \equiv 1 \pmod{11}$ and $N \equiv 3 \pmod{7}$: the candidates $1, 12, 23, 34, 45, \ldots$ leave remainders $1, 5, 2, 6, 3$ on division by 7, so $N = 45$.

(Option (A) is the remainder modulo 7 only, option (B) the remainder modulo 11 only.)

**Correct Option: (D)**

## Question 08

**English**

The value of $\displaystyle\sum_{0\le i<j\le 6}(j-i)\binom{6}{i}\binom{6}{j}$ is

(A) 1386
(B) 924
(C) 2772
(D) 5544

Only one option is correct.

**हिन्दी**

$\displaystyle\sum_{0\le i<j\le 6}(j-i)\binom{6}{i}\binom{6}{j}$ का मान है

(A) 1386
(B) 924
(C) 2772
(D) 5544

केवल एक विकल्प सही है।

**Solution**

Put $k=j-i$; the condition $0\le i<j\le 6$ means $1\le k\le 6$ and $0\le i\le 6-k$. So the double sum is $\sum_{k=1}^{6} k\,S_k$ with $S_k=\sum_{i=0}^{6-k}\binom{6}{i}\binom{6}{i+k}$.

Since $\binom{6}{i+k}=\binom{6}{6-k-i}$, $S_k$ is the coefficient of $x^{6-k}$ in $(1+x)^6(1+x)^6=(1+x)^{12}$, i.e. $S_k=\binom{12}{6-k}=\binom{12}{6+k}$.

Values: $\binom{12}{7}=792$, $\binom{12}{8}=495$, $\binom{12}{9}=220$, $\binom{12}{10}=66$, $\binom{12}{11}=12$, $\binom{12}{12}=1$.

Sum: $792+2\cdot 495+3\cdot 220+4\cdot 66+5\cdot 12+6\cdot 1=792+990+660+264+60+6=2772$.

Check by a closed form: $\sum_{k\ge 1}k\binom{2n}{n+k}=\dfrac{n}{2}\binom{2n}{n}$; with $n=6$, $3\cdot 924=2772$.

(Option (A) is half the answer and option (B) is $\binom{12}{6}$; option (D) double counts ordered pairs.)

**Correct Option: (C)**

## Question 09

**English**

For each positive integer $n$, let $c_n$ denote the coefficient of $x^n$ in the expansion of $\left(x^2 - \frac{3}{x}\right)^n$, $x \neq 0$ (with $c_n = 0$ if $x^n$ does not occur). Then the value of $\sum_{n=1}^{9} c_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

प्रत्येक धनात्मक पूर्णांक $n$ के लिए माना $c_n$, $\left(x^2 - \frac{3}{x}\right)^n$, $x \neq 0$, के प्रसार में $x^n$ का गुणांक है (यदि $x^n$ प्रसार में नहीं आता तो $c_n = 0$)। तब $\sum_{n=1}^{9} c_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

General term of $\left(x^2 - \frac{3}{x}\right)^n$: $T_{r+1} = \binom{n}{r}\left(x^2\right)^{n-r}\left(-\frac{3}{x}\right)^r = \binom{n}{r}(-3)^r x^{2n-3r}$, $0 \le r \le n$.

The term in $x^n$ needs $2n - 3r = n$, i.e. $r = \frac{n}{3}$. This is an integer in $[0, n]$ only when $3 \mid n$.

So for $n = 1, 2, 4, 5, 7, 8$ the power $x^n$ does not occur and $c_n = 0$.

$n = 3$, $r = 1$: $c_3 = \binom{3}{1}(-3) = -9$.

$n = 6$, $r = 2$: $c_6 = \binom{6}{2}(-3)^2 = 15 \cdot 9 = 135$.

$n = 9$, $r = 3$: $c_9 = \binom{9}{3}(-3)^3 = 84 \cdot (-27) = -2268$.

$\sum_{n=1}^{9} c_n = -9 + 135 - 2268 = -2142$.

**Final Answer: -2142**

## Question 10

**English**

Let $a$ be a real number such that the term independent of $x$ in the expansion of $(1+ax)^3\left(x+\frac{1}{x}\right)^5$, $x \neq 0$, is 100. Then the coefficient of $\frac{1}{x^2}$ in this expansion is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $a$ एक ऐसी वास्तविक संख्या है कि $(1+ax)^3\left(x+\frac{1}{x}\right)^5$, $x \neq 0$, के प्रसार में $x$ से स्वतंत्र पद 100 है। तब इस प्रसार में $\frac{1}{x^2}$ का गुणांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Second factor: $\left(x+\frac{1}{x}\right)^5 = \sum_{r=0}^{5}\binom{5}{r}x^{5-2r}$, with terms $x^5, 5x^3, 10x, 10x^{-1}, 5x^{-3}, x^{-5}$; every power is odd.

First factor: $(1+ax)^3 = 1+3ax+3a^2x^2+a^3x^3$.

A product term $x^j \cdot x^{5-2r}$ has an even exponent only when $j$ is odd, so only $3ax$ and $a^3x^3$ contribute to even powers such as $x^0$ and $x^{-2}$.

Term independent of $x$: $3ax \cdot 10x^{-1} + a^3x^3 \cdot 5x^{-3} = 30a + 5a^3$. Given $30a + 5a^3 = 100$, i.e. $a^3 + 6a - 20 = 0$.

$a = 2$ satisfies it $(8 + 12 - 20 = 0)$. Since $\frac{d}{da}(a^3+6a) = 3a^2+6 > 0$, the cubic has exactly one real root, so $a = 2$.

Coefficient of $x^{-2}$: $3ax \cdot 5x^{-3} + a^3x^3 \cdot x^{-5}$, i.e. $15a + a^3 = 30 + 8 = 38$.

**Final Answer: 38**

## Question 11

**English**

Let $p$ be a positive integer such that the expansion of $\left(x^p - \frac{1}{4x}\right)^6$, $x \neq 0$, contains a term independent of $x$ and also contains a term in $x^{24}$. Then the coefficient of $x^{24}$ in this expansion is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $p$ एक ऐसा धनात्मक पूर्णांक है कि $\left(x^p - \frac{1}{4x}\right)^6$, $x \neq 0$, के प्रसार में $x$ से स्वतंत्र एक पद है तथा $x^{24}$ वाला एक पद भी है। तब इस प्रसार में $x^{24}$ का गुणांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

General term: $T_{r+1} = \binom{6}{r}(x^p)^{6-r}\left(-\frac{1}{4x}\right)^r = \binom{6}{r}\left(-\frac{1}{4}\right)^r x^{6p-(p+1)r}$, $r = 0, 1, \ldots, 6$.

Term independent of $x$: $6p = (p+1)r$, so $r = \frac{6p}{p+1}$. Since $p$ and $p+1$ are coprime, $p+1$ must divide 6, giving $p+1 \in \{2, 3, 6\}$, i.e. $p \in \{1, 2, 5\}$ (then $r = 3, 4, 5$, all within $0 \leq r \leq 6$).

Term in $x^{24}$: the exponents are $6p$, $5p-1, \ldots$; the largest is $6p$, so $6p \geq 24$, i.e. $p \geq 4$. Of $1, 2, 5$ only $p = 5$ remains.

For $p = 5$ the exponent is $30 - 6r$, and $30 - 6r = 24$ gives $r = 1$, so $x^{24}$ does occur.

Its coefficient: $\binom{6}{1}\left(-\frac{1}{4}\right)^1 = -\frac{6}{4} = -1.5$.

**Final Answer: -1.50**

## Question 12

**English**

Let $n$ be the positive integer for which the greatest binomial coefficient in the expansion of $(1+y)^n$ is 35. In the expansion of $\left(2x+\frac{6}{5x}\right)^n$, $x \neq 0$, let $P$ and $Q$ be the two middle terms, where $P$ contains $x$ and $Q$ contains $\frac{1}{x}$. Then the value of $\frac{x^2Q}{P}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $n$ वह धनात्मक पूर्णांक है जिसके लिए $(1+y)^n$ के प्रसार में महत्तम द्विपद गुणांक 35 है। $\left(2x+\frac{6}{5x}\right)^n$, $x \neq 0$, के प्रसार में माना $P$ तथा $Q$ दो मध्य पद हैं, जहाँ $P$ में $x$ तथा $Q$ में $\frac{1}{x}$ है। तब $\frac{x^2Q}{P}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The greatest binomial coefficient of $(1+y)^n$ is the middle one, $\binom{n}{\lfloor n/2 \rfloor}$, which increases with $n$: $\binom{6}{3}=20$, $\binom{7}{3}=35$, $\binom{8}{4}=70$. So $n=7$.

For $n=7$ there are 8 terms, and the middle terms are $T_4$ and $T_5$. Their binomial coefficients $\binom{7}{3}$ and $\binom{7}{4}$ are equal, both being the greatest coefficient 35.

$T_4=\binom{7}{3}(2x)^4\left(\frac{6}{5x}\right)^3$ contains $x^{4-3}=x$, so $P=T_4$; $T_5=\binom{7}{4}(2x)^3\left(\frac{6}{5x}\right)^4$ contains $x^{3-4}=x^{-1}$, so $Q=T_5$.

Because the binomial coefficients are equal, $\frac{Q}{P}=\frac{(2x)^3(6/(5x))^4}{(2x)^4(6/(5x))^3}=\frac{6/(5x)}{2x}=\frac{3}{5x^2}$.

Hence $\frac{x^2Q}{P}=\frac{3}{5}=0.6$, independent of $x$.

(Check: $P=35\cdot 16\cdot\frac{216}{125}x$, $Q=35\cdot 8\cdot\frac{1296}{625}x^{-1}$, and $\frac{x^2Q}{P}=\frac{8\cdot 1296\cdot 125}{16\cdot 216\cdot 625}=\frac{3}{5}$.)

**Final Answer: 0.60**

## Question 13

**English**

Let $n$ be the positive integer for which the expansion of $\left(\sqrt{10}+\sqrt{7}\right)^n$ by the binomial theorem has exactly 4 rational terms. Let $I$ and $f$ be the integral part and the fractional part of $\left(\sqrt{10}+\sqrt{7}\right)^n$, so that $I$ is an integer and $0 \le f < 1$. Then the value of $(I+f)(1-f)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $n$ वह धनात्मक पूर्णांक है जिसके लिए द्विपद प्रमेय द्वारा $\left(\sqrt{10}+\sqrt{7}\right)^n$ के प्रसार में ठीक $4$ परिमेय पद हैं। माना $I$ तथा $f$ क्रमशः $\left(\sqrt{10}+\sqrt{7}\right)^n$ के पूर्णांक भाग तथा भिन्नात्मक भाग हैं, अर्थात् $I$ एक पूर्णांक है तथा $0 \le f < 1$ है। तब $(I+f)(1-f)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

General term: $T_{r+1} = \binom{n}{r}\left(\sqrt{10}\right)^{n-r}\left(\sqrt{7}\right)^r = \binom{n}{r}10^{\frac{n-r}{2}}7^{\frac{r}{2}}$.

It is rational exactly when $\frac{n-r}{2}$ and $\frac{r}{2}$ are integers, i.e. $r$ and $n-r$ are both even. For odd $n$ this is impossible (no rational term). For even $n$ the even $r = 0, 2, \ldots, n$ work: $\dfrac{n}{2}+1$ rational terms.

Exactly 4 rational terms: $\dfrac{n}{2}+1 = 4$, so $n = 6$.

Let $g = \left(\sqrt{10}-\sqrt{7}\right)^6$. As $\sqrt{10}-\sqrt{7} = \dfrac{3}{\sqrt{10}+\sqrt{7}}$ and $\sqrt{10}+\sqrt{7} > 3$, we get $0 < \sqrt{10}-\sqrt{7} < 1$, hence $0 < g < 1$.

In $\left(\sqrt{10}+\sqrt{7}\right)^6 + \left(\sqrt{10}-\sqrt{7}\right)^6$ the odd-$r$ terms cancel and the even-$r$ (rational) terms double, so $I+f+g$ is an integer.

Then $f+g$ is an integer with $0 < f+g < 2$, so $f+g = 1$ and $1-f = g$.

Therefore $(I+f)(1-f) = \left(\sqrt{10}+\sqrt{7}\right)^6\left(\sqrt{10}-\sqrt{7}\right)^6 = (10-7)^6 = 3^6 = 729$.

(Numerically $(\sqrt{10}+\sqrt{7})^6 \approx 38\,385.98$ and $g \approx 0.019$, consistent with $f \approx 0.981$.)

**Final Answer: 729**

## Question 14

**English**

Let $S$ be the set of all real numbers $x > 0$ for which no term in the expansion of $(2 + x)^8$ is numerically greater than the term having the greatest binomial coefficient. If $\alpha$ and $\beta$ are the least and the greatest elements of $S$ respectively, then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $S$ उन सभी वास्तविक संख्याओं $x > 0$ का समुच्चय है जिनके लिए $(2 + x)^8$ के प्रसार का कोई भी पद, महत्तम द्विपद गुणांक वाले पद से संख्यात्मक रूप से बड़ा नहीं है। यदि $\alpha$ तथा $\beta$ क्रमशः $S$ के न्यूनतम तथा अधिकतम अवयव हैं, तो $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Terms: $T_{r+1} = \binom{8}{r} 2^{8-r} x^r$, $r = 0, \ldots, 8$ (all positive for $x > 0$). The greatest binomial coefficient is $\binom{8}{4} = 70$, in $T_5$.

Ratio of consecutive terms: $\dfrac{T_{r+1}}{T_r} = \dfrac{\binom{8}{r}}{\binom{8}{r-1}} \cdot \dfrac{x}{2} = \dfrac{9-r}{r} \cdot \dfrac{x}{2}$, which decreases as $r$ increases. So the terms rise and then fall, and $T_5$ is not exceeded by any term exactly when $T_5 \geq T_4$ and $T_5 \geq T_6$.

$T_5 \geq T_4$: $\dfrac{5}{4} \cdot \dfrac{x}{2} \geq 1 \iff x \geq \dfrac{8}{5}$.

$T_6 \leq T_5$: $\dfrac{4}{5} \cdot \dfrac{x}{2} \leq 1 \iff x \leq \dfrac{5}{2}$.

So $S = \left[\dfrac{8}{5}, \dfrac{5}{2}\right]$; at the ends $T_5$ ties with $T_4$ or $T_6$, which is allowed ('no term is greater').

$\alpha = \dfrac{8}{5}$, $\beta = \dfrac{5}{2}$, $\alpha + \beta = \dfrac{41}{10} = 4.1$.

**Final Answer: 4.10**

## Question 15

**English**

Let $n$ be the positive integer for which the greatest binomial coefficient in the expansion of $(1+y)^n$ is 70. In the expansion of $(1-3x)^n$ in ascending powers of $x$, let $T$ be the numerically greatest term and $M$ be the middle term, both evaluated at $x=1$. Then the value of $\dfrac{T}{M}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $n$ वह धनात्मक पूर्णांक है जिसके लिए $(1+y)^n$ के प्रसार में महत्तम द्विपद गुणांक 70 है। $(1-3x)^n$ के $x$ की आरोही घातों में प्रसार में माना $T$ संख्यात्मक रूप से महत्तम पद तथा $M$ मध्य पद है, दोनों का मान $x=1$ पर लिया गया है। तब $\dfrac{T}{M}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The greatest binomial coefficient of $(1+y)^n$ is the middle one and grows with $n$: $\binom{7}{3}=35$, $\binom{8}{4}=70$, $\binom{9}{4}=126$. So $n=8$.

Terms of $(1-3x)^8$: $T_{r+1}=\binom{8}{r}(-3x)^r$. At $x=1$, $|T_{r+1}|=\binom{8}{r}3^r$.

$\dfrac{|T_{r+1}|}{|T_r|}=\dfrac{9-r}{r}\cdot 3$. This is $\geq 1$ iff $3(9-r)\geq r$, i.e. $r\leq \dfrac{27}{4}=6.75$. So $|T_r|$ increases up to $r=6$, i.e. up to $T_7$, and decreases after it; $T_7$ is the unique numerically greatest term.

$T_7=\binom{8}{6}(-3)^6=28\cdot 729=20\,412$ (positive, as the power of $-3$ is even; $|T_6|=13\,608$, $|T_8|=17\,496$).

There are 9 terms, so the middle term is $T_5=\binom{8}{4}(-3)^4=70\cdot 81=5\,670$.

$\dfrac{T}{M}=\dfrac{20\,412}{5\,670}=\dfrac{18}{5}=3.6$.

**Final Answer: 3.60**

## Question 16

**English**

For a real number $t$, let $\{t\}$ denote the fractional part of $t$. Then the value of $\left\{\frac{3^{2026}}{25}\right\} - \left\{\frac{7^{2026}}{25}\right\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्या $t$ के लिए माना $\{t\}$, $t$ का भिन्नात्मक भाग दर्शाता है। तब $\left\{\frac{3^{2026}}{25}\right\} - \left\{\frac{7^{2026}}{25}\right\}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

If $N = 25q + s$ with $0 \le s < 25$, then $\left\{\frac{N}{25}\right\} = \frac{s}{25}$. So only the remainders modulo 25 are needed.

$7^{2026} = (7^2)^{1013} = (50-1)^{1013} = \sum_{k=0}^{1013} \binom{1013}{k} 50^k (-1)^{1013-k}$. Every term with $k \ge 1$ is a multiple of 50, so $7^{2026} = 50m - 1$ for an integer $m$, i.e. $7^{2026} \equiv -1 \equiv 24 \pmod{25}$.

$3^{2026} = (3^2)^{1013} = (10-1)^{1013} = \sum_k \binom{1013}{k} 10^k (-1)^{1013-k}$. Terms with $k \ge 2$ contain 100, a multiple of 25. So $3^{2026} \equiv (-1) + 1013 \cdot 10 = 10129 \pmod{25}$, and $10129 = 25 \cdot 405 + 4$, giving remainder 4.

Hence $\left\{\frac{3^{2026}}{25}\right\} = \frac{4}{25} = 0.16$ and $\left\{\frac{7^{2026}}{25}\right\} = \frac{24}{25} = 0.96$.

The required value is $0.16 - 0.96 = -0.8$.

**Final Answer: -0.80**

## Question 17

**English**

The number of positive integers $n \le 150$ for which $11^n + 9^n$ is divisible by 100 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

ऐसे धनात्मक पूर्णांकों $n \le 150$ की संख्या, जिनके लिए $11^n + 9^n$, 100 से विभाज्य है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Expand: $11^n = (1+10)^n = 1 + 10n + \binom{n}{2}100 + \cdots \equiv 1 + 10n \pmod{100}$; every later term contains $10^2$.

$9^n = (10-1)^n = (-1)^n + n \cdot 10 \cdot (-1)^{n-1} + \binom{n}{2}100(-1)^{n-2} + \cdots \equiv (-1)^n + 10n(-1)^{n-1} \pmod{100}$.

For even $n$: $9^n \equiv 1 - 10n$, so $11^n + 9^n \equiv 2 \pmod{100}$. The last two digits are 02; never divisible by 100.

For odd $n$: $9^n \equiv -1 + 10n$, so $11^n + 9^n \equiv 20n \pmod{100}$. This is 0 modulo 100 iff $100 \mid 20n$, i.e. $5 \mid n$.

So the condition is: $n$ odd and $5 \mid n$, i.e. $n \equiv 5 \pmod{10}$. In $1 \le n \le 150$: $n = 5, 15, 25, \ldots, 145$, which is $\frac{145-5}{10} + 1 = 15$ values.

(Check $n = 5$: $11^5 + 9^5 = 161051 + 59049 = 220100$.)

**Final Answer: 15**

## Question 18

**English**

Let $N$ be the number formed by the last three digits of $3^{4052}$, so that $0 \le N \le 999$ and $3^{4052} - N$ is divisible by 1000. Then the value of $N$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $N$, $3^{4052}$ के अंतिम तीन अंकों से बनी संख्या है, अर्थात् $0 \le N \le 999$ है तथा $3^{4052} - N$, 1000 से विभाज्य है। तब $N$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$3^{4052} = (3^2)^{2026} = 9^{2026} = (10-1)^{2026} = \sum_{k=0}^{2026} \binom{2026}{k} 10^k (-1)^{2026-k}$.

For $k \ge 3$ each term is a multiple of 1000, so modulo 1000 only $k = 0, 1, 2$ remain (with signs $+, -, +$ since 2026 is even):

$3^{4052} \equiv 1 - \binom{2026}{1}10 + \binom{2026}{2}100 \pmod{1000}$.

$\binom{2026}{1}10 = 20260 \equiv 260$, so $-20260 \equiv -260 \equiv 740$.

$\binom{2026}{2} = \dfrac{2026 \cdot 2025}{2} = 1013 \cdot 2025 = 2\,051\,325$, and $2\,051\,325 \cdot 100 = 205\,132\,500 \equiv 500 \pmod{1000}$.

Total: $1 + 740 + 500 = 1241 \equiv 241 \pmod{1000}$. So $N = 241$.

**Final Answer: 241**

## Question 19

**English**

Let $[t]$ denote the greatest integer less than or equal to $t$. Then the value of $\left[\left(1-\sqrt{2}\right)^{-7}\right]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $[t]$, $t$ से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\left[\left(1-\sqrt{2}\right)^{-7}\right]$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Since $(1-\sqrt{2})(1+\sqrt{2}) = 1-2 = -1$, $(1-\sqrt{2})^{-1} = -(1+\sqrt{2})$ and so $(1-\sqrt{2})^{-7} = (-1)^7(1+\sqrt{2})^7 = -(1+\sqrt{2})^7$.

In $(1+\sqrt{2})^7+(1-\sqrt{2})^7$ the odd powers of $\sqrt{2}$ cancel and the even ones double: $2\left[\binom{7}{0} + \binom{7}{2}(\sqrt{2})^2 + \binom{7}{4}(\sqrt{2})^4 + \binom{7}{6}(\right.$
$2(1 + 42 + 140 + 56) = 478$.

$1 - \sqrt{2} = -(\sqrt{2} - 1)$ with $0 < \sqrt{2} - 1 < 1$, so $(1-\sqrt{2})^7 = -(\sqrt{2}-1)^7$ lies in $(-1, 0)$.

Hence $(1+\sqrt{2})^7 = 478 + (\sqrt{2}-1)^7$, a number slightly larger than 478 (about 478.0021).

Therefore $(1-\sqrt{2})^{-7} = -478 - (\sqrt{2}-1)^7$, which lies strictly between $-479$ and $-478$, and its greatest integer is $-479$.

(Taking the integer part of $(1+\sqrt{2})^7$ and only changing the sign would give $-478$, which is wrong for a negative non-integer.)

**Final Answer: -479**

## Question 20

**English**

The value of $\displaystyle\sum_{r=0}^{15}(-1)^r\,\frac{r^2+3}{r+1}\binom{15}{r}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$\displaystyle\sum_{r=0}^{15}(-1)^r\,\frac{r^2+3}{r+1}\binom{15}{r}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Divide: $r^2+3=(r+1)(r-1)+4$, so $\dfrac{r^2+3}{r+1}=r-1+\dfrac{4}{r+1}$.

The sum splits as $\sum(-1)^r r\binom{15}{r}-\sum(-1)^r\binom{15}{r}+4\sum(-1)^r\dfrac{\binom{15}{r}}{r+1}$.

$\sum_{r=0}^{15}(-1)^r\binom{15}{r}=(1-1)^{15}=0$.

Differentiate $(1-x)^{15}=\sum(-1)^r\binom{15}{r}x^r$: $-15(1-x)^{14}=\sum(-1)^r r\binom{15}{r}x^{r-1}$; at $x=1$ this gives $\sum(-1)^r r\binom{15}{r}=0$.

Integrate $(1-x)^{15}$ from 0 to 1: $\dfrac{1}{16}=\sum(-1)^r\binom{15}{r}\dfrac{1}{r+1}$.

Therefore the sum is $0-0+\dfrac{4}{16}=\dfrac{1}{4}=0.25$.

**Final Answer: 0.25**

## Question 21

**English**

For a real number $t$, let $\{t\}$ denote the fractional part of $t$. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \sum_{r=0}^{9} \binom{9}{r} (-2)^{9-r} \{x\}^r$. Then the value of $\int_0^{10} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्या $t$ के लिए माना $\{t\}$, $t$ का भिन्नात्मक भाग दर्शाता है। माना $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sum_{r=0}^{9} \binom{9}{r} (-2)^{9-r} \{x\}^r$ द्वारा परिभाषित है। तब $\int_0^{10} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

By the binomial theorem, $\sum_{r=0}^{9} \binom{9}{r} \{x\}^r (-2)^{9-r} = (\{x\} - 2)^9$, so $f(x) = (\{x\} - 2)^9$. Since $\{x+1\} = \{x\}$, $f$ has period 1.

A tempting antiderivative is $F(x) = \dfrac{(\{x\} - 2)^{10}}{10}$: on each open interval $(k, k+1)$, $\{x\} = x - k$ and $F'(x) = f(x)$. But $F$ is not continuous at the integers ($F(k^-) = \dfrac{1}{10}$, $F(k) = \dfrac{1024}{10}$), so $F(10) - F(0) = 0$ is not the value of the integral.

Split $[0, 10]$ into $[k, k+1]$, $k = 0, \ldots, 9$. On $[k, k+1)$, $\{x\} = x - k$, and with $t = x - k$, each piece equals $\int_0^1 (t-2)^9 dt$.

Term by term (integration technique): $\int_0^1 (t-2)^9 dt = \sum_{r=0}^{9} \binom{9}{r} (-2)^{9-r} \int_0^1 t^r\,dt = \sum_{r=0}^{9} \binom{9}{r} \dfrac{(-2)^{9-r}}{r+1}$.

This sum is evaluated by the continuous antiderivative on $[0, 1]$: $\left[\dfrac{(t-2)^{10}}{10}\right]_0^1 = \dfrac{(-1)^{10} - (-2)^{10}}{10} = \dfrac{1 - 1024}{10} = -102.3$.

Hence $\int_0^{10} f(x)\,dx = 10 \cdot (-102.3) = -1023$.

**Final Answer: -1023**

## Question 22

**English**

Let $A = \sum_{k=0}^{9} (-1)^k \binom{28}{3k}$ and $B = \sum_{k=0}^{8} (-1)^k \binom{28}{3k+2}$. Then the value of $\frac{A}{B}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $A = \sum_{k=0}^{9} (-1)^k \binom{28}{3k}$ तथा $B = \sum_{k=0}^{8} (-1)^k \binom{28}{3k+2}$ है। तब $\frac{A}{B}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The roots of $z^3 = -1$ are $z = -1$, $\omega = e^{i\pi/3}$ and $\bar{\omega} = e^{-i\pi/3}$, i.e. $z = -\zeta$ where $\zeta^3 = 1$. For an integer $s$: $\sum_{z^3=-1} z^s = (-1)^s \sum_{\zeta^3=1} \zeta^s$, which is $3(-1)^s = 3(-1)^{s/3}$ when $3 \mid s$ (as $s$ and $s/3$ have the same parity) and 0 otherwise.

Hence $\sum_{z^3=-1} z^{-j}(1+z)^{28} = \sum_m \binom{28}{m} \sum_z z^{m-j} = 3 \sum_{m \equiv j\,(3)} (-1)^{(m-j)/3} \binom{28}{m}$. With $j = 0$ this is $3A$; with $j = 2$ ($m = 3k+2$) it is $3B$.

The root $z = -1$ contributes 0 since $1 + z = 0$. For $z = \omega$: $1 + \omega = \frac{3}{2} + \frac{\sqrt{3}}{2} i = \sqrt{3} e^{i\pi/6}$, so $(1+\omega)^{28} = 3^{14} e^{i28\pi/6} = 3^{14} e^{i14\pi/3} = 3^{14} e^{i2\pi/3}$. The root $\bar{\omega}$ gives the conjugate.

$3A = 2\,\mathrm{Re}\left[3^{14} e^{i2\pi/3}\right] = 2 \cdot 3^{14} \left(-\frac{1}{2}\right) = -3^{14}$, so $A = -3^{13} = -1\,594\,323$.

$3B = 2\,\mathrm{Re}\left[\omega^{-2} 3^{14} e^{i2\pi/3}\right] = 2 \cdot 3^{14}\,\mathrm{Re}\left[e^{-i2\pi/3} e^{i2\pi/3}\right] = 2 \cdot 3^{14}$, so $B = 2 \cdot 3^{13} = 3\,188\,646$.

Therefore $\frac{A}{B} = -\frac{1}{2} = -0.5$ (both values agree with direct computation).

**Final Answer: -0.50**

## Question 23

**English**

The sum $\sum_{k=3}^{14} \frac{\binom{k}{2}}{\binom{k}{3}\binom{k+1}{3}}$ can be written as $1 - \frac{1}{N}$ for a positive integer $N$. Then the value of $N$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

योग $\sum_{k=3}^{14} \frac{\binom{k}{2}}{\binom{k}{3}\binom{k+1}{3}}$ को किसी धनात्मक पूर्णांक $N$ के लिए $1-\frac{1}{N}$ के रूप में लिखा जा सकता है। तब $N$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Pascal's rule $\binom{k+1}{3} = \binom{k}{3} + \binom{k}{2}$ gives $\binom{k}{2} = \binom{k+1}{3} - \binom{k}{3}$.

Hence each term is $\frac{\binom{k+1}{3} - \binom{k}{3}}{\binom{k}{3}\binom{k+1}{3}} = \frac{1}{\binom{k}{3}} - \frac{1}{\binom{k+1}{3}}$ (valid for $k \geq 3$, where $\binom{k}{3} \neq 0$).

Summing from $k = 3$ to 14, the inner terms cancel: $\frac{1}{\binom{3}{3}} - \frac{1}{\binom{15}{3}} = 1 - \frac{1}{455}$.

$\binom{15}{3} = \frac{15 \cdot 14 \cdot 13}{6} = 455$, so $N = 455$ (the representation $1 - \frac{1}{N}$ is unique).

**Final Answer: 455**

## Question 24

**English**

The value of $\sum_{n=1}^{\infty} \frac{n}{\binom{n+6}{6}}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$\sum_{n=1}^{\infty} \frac{n}{\binom{n+6}{6}}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Write $\frac{1}{\binom{n+k}{k}} = \frac{k!\,n!}{(n+k)!}$.

Factorial telescoping: $\frac{n!}{(n+k-1)!} - \frac{(n+1)!}{(n+k)!} = \frac{n!}{(n+k)!}\left[(n+k) - (n+1)\right] = (k-1)\frac{n!}{(n+k)!}$.

So $\sum_{n=1}^{\infty} \frac{n!}{(n+k)!} = \frac{1}{k-1} \cdot \frac{1!}{k!}$ (the terms $\frac{(n+1)!}{(n+k)!} \to 0$ for $k \geq 2$), and therefore $\sum_{n=1}^{\infty} \frac{1}{\binom{n+k}{k}} = \frac{k!}{(k-1)\,k!} = \frac{1}{k-1}$.

Now split $n = (n+6) - 6$: $\frac{n}{\binom{n+6}{6}} = \frac{n+6}{\binom{n+6}{6}} - \frac{6}{\binom{n+6}{6}}$, and $\frac{n+6}{\binom{n+6}{6}} = \frac{6!\,n!\,(n+6)}{(n+6)!} = \frac{6 \cdot 5!\,n!}{(n+5)!} = \frac{6}{\binom{n+5}{5}}$.

Hence the sum is $6\sum_{n\geq 1}\frac{1}{\binom{n+5}{5}} - 6\sum_{n\geq 1}\frac{1}{\binom{n+6}{6}} = 6\cdot\frac{1}{4} - 6\cdot\frac{1}{5} = \frac{6}{20} = \frac{3}{10}$.

(Both series converge, since their terms behave like $n^{-5}$ and $n^{-6}$; the numerical partial sums approach 0.3.)

**Final Answer: 0.30**

## Question 25

**English**

The coefficient of $x^3$ in the expansion of $\left(1+\frac{x}{2}+\frac{x^2}{4}\right)^5$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$\left(1+\frac{x}{2}+\frac{x^2}{4}\right)^5$ के प्रसार में $x^3$ का गुणांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Multinomial theorem: $(p+q+s)^5 = \sum \frac{5!}{a!\,b!\,c!}p^a q^b s^c$ over $a+b+c=5$. With $p=1$, $q=\frac{x}{2}$, $s=\frac{x^2}{4}$ the general term is $\frac{5!}{a!\,b!\,c!}\cdot\frac{1}{2^b 4^c}x^{b+2c}$.

For $x^3$: $b+2c=3$ with $a=5-b-c\geq 0$. So $c=0, b=3, a=2$ or $c=1, b=1, a=3$.

$(2,3,0)$: $\frac{120}{2\cdot 6}\cdot\frac{1}{8} = \frac{10}{8}$.

$(3,1,1)$: $\frac{120}{6}\cdot\frac{1}{2}\cdot\frac{1}{4} = \frac{20}{8}$.

Total: $\frac{30}{8} = \frac{15}{4} = 3.75$.

(Check: $1+y+y^2 = \frac{1-y^3}{1-y}$ with $y=\frac{x}{2}$, so the expression is $(1-y^3)^5(1-y)^{-5}$; the $y^3$ coefficient is $\binom{7}{3}-5 = 30$, and $30\cdot\frac{1}{8} = \frac{15}{4}$.)

**Final Answer: 3.75**

## Question 26

**English**

When $\left(1+x+x^3\right)^7$ is expanded and the like terms are combined, the number of terms in the resulting expression is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

जब $\left(1+x+x^3\right)^7$ का प्रसार करके समान पदों को मिलाया जाता है, तो प्राप्त व्यंजक में पदों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

By the multinomial theorem, $(1+x+x^3)^7 = \sum \dfrac{7!}{a!\,b!\,c!}x^{b+3c}$ over non-negative $a+b+c=7$. There are $\binom{9}{2} = 36$ triples, but triples with the same $b+3c$ combine, so the number of terms is the number of distinct exponents $b+3c$.

For a fixed $c$ ($0 \le c \le 7$), $b$ runs over $0, \ldots, 7-c$, so the exponent runs over the interval $3c, 3c+1, \ldots, 7+2c$.

$c=0$: 0 to 7; $c=1$: 3 to 9; $c=2$: 6 to 11; $c=3$: 9 to 13; $c=4$: 12 to 15; $c=5$: 15 to 17; $c=6$: 18 to 19; $c=7$: 21.

Together these cover every integer from 0 to 19, and 21. The exponent 20 would need $c=6, b=2$ ($b+c=8$) or $c=5, b=5$ ($b+c=10$), both impossible.

All multinomial coefficients are positive, so no combined coefficient vanishes. Number of terms $= 20+1 = 21$.

(The answers 22 (all powers 0 to 21) and 36 (the multinomial count) are the two traps.)

**Final Answer: 21**

## Question 27

**English**

Let $f : \left(-\frac{1}{2}, \frac{1}{2}\right) \to \mathbb{R}$ be defined by $f(x) = \frac{e^x}{\sqrt{1+2x}}$, and let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for all $x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$. Then the value of $a_4$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना $f : \left(-\frac{1}{2}, \frac{1}{2}\right) \to \mathbb{R}$, $f(x) = \frac{e^x}{\sqrt{1+2x}}$ द्वारा परिभाषित है तथा सभी $x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$ के लिए $f(x) = \sum_{n=0}^{\infty} a_n x^n$ है। तब $a_4$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Binomial theorem for index $-\frac{1}{2}$ (valid as $|2x| < 1$): the coefficient of $x^k$ in $(1+2x)^{-\frac{1}{2}}$ is $\binom{-1/2}{k} 2^k$.

$k = 1$: $-\frac{1}{2} \cdot 2 = -1$; $k = 2$: $\frac{(-\frac{1}{2})(-\frac{3}{2})}{2} \cdot 4 = \frac{3}{2}$; $k = 3$: $\frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{6} \cdot 8 = -\frac{5}{2}$; $k = 4$: $\frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})(-\frac{7}{2})}{24} \cdot 16 = \frac{35}{8}$.

Exponential series: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$

The power series of $f$ is the product of the two series, so $a_4 = \sum_{k=0}^{4} \left(\text{coefficient of } x^k\right) \cdot \frac{1}{(4-k)!}$.

$$a_4 = 1 \cdot \frac{1}{24} + (-1) \cdot \frac{1}{6} + \frac{3}{2} \cdot \frac{1}{2} + \left(-\frac{5}{2}\right) \cdot 1 + \frac{35}{8} \cdot 1 = \frac{1 - 4 + 18 - 60 + 105}{24} = \frac{60}{24} = \frac{5}{2}.$$

(Lower coefficients for a check: $a_1 = 0$, $a_2 = 1$, $a_3 = -\frac{4}{3}$.)

**Final Answer: 2.50**

## Question 28

**English**

Let $a$ be a real number such that $\lim\limits_{x\to 0} \dfrac{(1+ax)^{\frac{1}{8}} - (1-x)^{-\frac{1}{2}}}{x^2}$ exists and is a real number $L$. Then the value of $L$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $a$ एक ऐसी वास्तविक संख्या है कि $\lim\limits_{x\to 0} \dfrac{(1+ax)^{\frac{1}{8}} - (1-x)^{-\frac{1}{2}}}{x^2}$ का अस्तित्व है तथा यह एक वास्तविक संख्या $L$ है। तब $L$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For $|x|$ small, the binomial theorem for fractional and negative indices gives $(1+ax)^{\frac{1}{8}} = 1 + \dfrac{1}{8}ax + \dfrac{\frac{1}{8}\cdot\left(-\frac{7}{8}\right)}{2!}a^2x^2 + O(x^3) = 1 + \dfrac{a}{8}x - \dfrac{7a^2}{128}x^2 + O(x^3)$.

$(1-x)^{-\frac{1}{2}} = 1 + \left(-\dfrac{1}{2}\right)(-x) + \dfrac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}x^2 + O(x^3) = 1 + \dfrac{x}{2} + \dfrac{3}{8}x^2 + O(x^3)$.

Numerator $= \left(\dfrac{a}{8} - \dfrac{1}{2}\right)x + \left(-\dfrac{7a^2}{128} - \dfrac{3}{8}\right)x^2 + O(x^3)$.

Dividing by $x^2$, the limit is finite only if the coefficient of $x$ is 0: $\dfrac{a}{8} = \dfrac{1}{2}$, so $a = 4$ (for any other $a$ the quotient is unbounded).

Then $L = -\dfrac{7\cdot 16}{128} - \dfrac{3}{8} = -\dfrac{7}{8} - \dfrac{3}{8} = -\dfrac{10}{8} = -\dfrac{5}{4} = -1.25$.

**Final Answer: -1.25**

## Question 29

**English**

The value of $\lim\limits_{x\to 0}\dfrac{(1+x)^{2x}-(1-x)^{-2x}}{x^3}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$\lim\limits_{x\to 0}\dfrac{(1+x)^{2x}-(1-x)^{-2x}}{x^3}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For $0<|x|<1$ both powers are defined: $(1+x)^{2x}=e^{2x\log_e(1+x)}$ and $(1-x)^{-2x}=e^{-2x\log_e(1-x)}$.

Logarithmic series: $\log_e(1+x)=x-\dfrac{x^2}{2}+O(x^3)$, so $2x\log_e(1+x)=2x^2-x^3+O(x^4)$.

$\log_e(1-x)=-x-\dfrac{x^2}{2}+O(x^3)$, so $-2x\log_e(1-x)=2x^2+x^3+O(x^4)$.

Exponential series: $e^u=1+u+\dfrac{u^2}{2}+\ldots$; here $u=O(x^2)$, so $u^2=O(x^4)$ and $e^u=1+u+O(x^4)$.

Hence $(1+x)^{2x}=1+2x^2-x^3+O(x^4)$ and $(1-x)^{-2x}=1+2x^2+x^3+O(x^4)$.

Their difference is $-2x^3+O(x^4)$, and the limit is $-2$.

**Final Answer: -2**

## Question 30

**English**

The remainder when $1^{40}+2^{40}+3^{40}+\cdots+100^{40}$ is divided by 41 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$1^{40}+2^{40}+3^{40}+\cdots+100^{40}$ को 41 से भाग देने पर शेषफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

41 is prime, so by Fermat's little theorem $k^{40}\equiv 1 \pmod{41}$ for every integer $k$ not divisible by 41.

Among $1,2,\ldots,100$ the multiples of 41 are 41 and 82; for these $k^{40}\equiv 0 \pmod{41}$.

The remaining $100-2=98$ terms each leave remainder 1, so the sum is $\equiv 98 \pmod{41}$.

$98=2\cdot 41+16$, so the remainder is 16.

(Forgetting the two multiples of 41 gives $100\equiv 18$.)

**Final Answer: 16**

## Question 31

**English**

For a real number $t$, let $\{t\}$ denote the fractional part of $t$. Then the value of $\left\{\dfrac{13^{2027}}{50}\right\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्या $t$ के लिए माना $\{t\}$, $t$ का भिन्नात्मक भाग दर्शाता है। तब $\left\{\dfrac{13^{2027}}{50}\right\}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

If $13^{2027} = 50q + s$ with $0 \le s < 50$, then the fractional part is $\dfrac{s}{50}$, so we need $13^{2027} \bmod 50$.

13 and 50 are coprime and $\phi(50) = \phi(2)\phi(25) = 1 \cdot 20 = 20$. By Euler's theorem $13^{20} \equiv 1 \pmod{50}$.

$2027 = 20 \cdot 101 + 7$, so $13^{2027} = (13^{20})^{101} \cdot 13^7 \equiv 13^7 \pmod{50}$.

$13^2 = 169 \equiv 19$; $13^4 \equiv 19^2 = 361 \equiv 11$; $13^6 \equiv 11 \cdot 19 = 209 \equiv 9$; $13^7 \equiv 9 \cdot 13 = 117 \equiv 17 \pmod{50}$.

Hence $\left\{\dfrac{13^{2027}}{50}\right\} = \dfrac{17}{50} = 0.34$.

**Final Answer: 0.34**

## Question 32

**English**

The value of $\displaystyle\sum_{0 \le i < j \le 6} \binom{6}{j}\binom{j}{i}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

$\displaystyle\sum_{0 \le i < j \le 6} \binom{6}{j}\binom{j}{i}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

The summation set is all pairs with $0 \le i < j \le 6$. Fix $j$ and sum over $i = 0, 1, \ldots, j-1$ first.

$\displaystyle\sum_{i=0}^{j-1} \binom{j}{i} = \sum_{i=0}^{j} \binom{j}{i} - \binom{j}{j} = 2^j - 1$ (for $j = 0$ the inner sum is empty and $2^0 - 1 = 0$ agrees).

So the double sum is $\displaystyle\sum_{j=0}^{6} \binom{6}{j}\left(2^j - 1\right) = \sum_{j=0}^{6} \binom{6}{j} 2^j - \sum_{j=0}^{6} \binom{6}{j}$.

By the binomial theorem, $\sum \binom{6}{j} 2^j = (1+2)^6 = 729$ and $\sum \binom{6}{j} = 2^6 = 64$.

Value $= 729 - 64 = 665$.

(Counting interpretation: choose a subset $J$ of $\{1, \ldots, 6\}$ and a subset of $J$ of smaller size: $3^6$ nested pairs minus the $2^6$ with equality.)

**Final Answer: 665**